Team: TEAM NUMBER REQUIRED

Problem B

Predicting a Penalty Kick from 0.10 Seconds of Motion

Full-flight inverse constraints and compatible trajectories

Abstract

Two position measurements do not uniquely identify a penalty kick when aerodynamic conditions are unknown. We apply gravity, quadratic drag relative to steady lateral wind, and an effective Magnus term from release throughout the flight. For prescribed wind, air density and drag coefficient, bounded nonlinear least squares jointly estimates initial velocity and two transverse-force coefficients. The six observation equations are relaxed to explicit componentwise tolerances. A grid of 231 parameter profiles yields 107 compatible representatives at a chosen 5 mm tolerance; their goal-plane coordinates span $y \in [2.5342, 2.6812]$ m and $z \in [1.9451, 1.9867]$ m, with two acceptable shots differing by 0.1513 m. This family is a sampled inner subset of feasible solutions, not an exhaustive uncertainty envelope or a probability distribution. Density and drag are structurally confounded through their product. A calm-air conditional reference remains $(11, 2.60, 1.97)$ m at 0.401 s; the observations alone do not define an unconditional most-likely point. We separately retain a 1 cm Gaussian-noise diagnostic, which gives approximately 0.97 m marginal 95% half-widths and must not be confused with the bounded-error family's spread. Analytical limiting cases, integration refinement, independent Python calculations and synthetic tests establish numerical consistency. No independent experimental flight is available. The method returns a conditional point, compatible full-flight trajectories, observation residuals and terminal-event classifications, while making its physical and statistical assumptions explicit.

1 Problem interpretation and deliverables

We use the coordinates and observations specified in Problem B [1]. The initial ball centre is $\mathbf{r}_0 = (0, 0, 0.11)$ m; the two subsequent observations are

$$\mathbf{r}_1 = (1.50, 0.45, 0.44) \text{ m}, \qquad \mathbf{r}_2 = (2.96, 0.86, 0.74) \text{ m}, \tag{1}$$

at $t_1 = 0.05$ s and $t_2 = 0.10$ s. Our primary prediction concerns the ball centre at its first forward crossing of $x = 11$ m, following the problem's plane convention. A separate finite-radius check assesses passage through the opening. The model does not predict the goalkeeper's response or collisions with the posts.

The requested general method must accept measured positions, estimate a trajectory, propagate uncertainty, and reject trajectories that hit the ground first or fail to reach the plane. We implement each stage explicitly. All reported uncertainty is conditional on stated noise and force assumptions; neither is fully specified by the question.

1.1 Objectives and solution workflow

We seek three outputs: a conditional point forecast, a set of physically admissible trajectories consistent with the observations, and diagnostics that distinguish observation mismatch from numerical failure. Figure 1 connects these outputs to the physical model and inverse calculation. The central difficulty is coupling: changing an environmental profile requires re-estimating launch conditions rather than propagating a fixed launch under every profile.

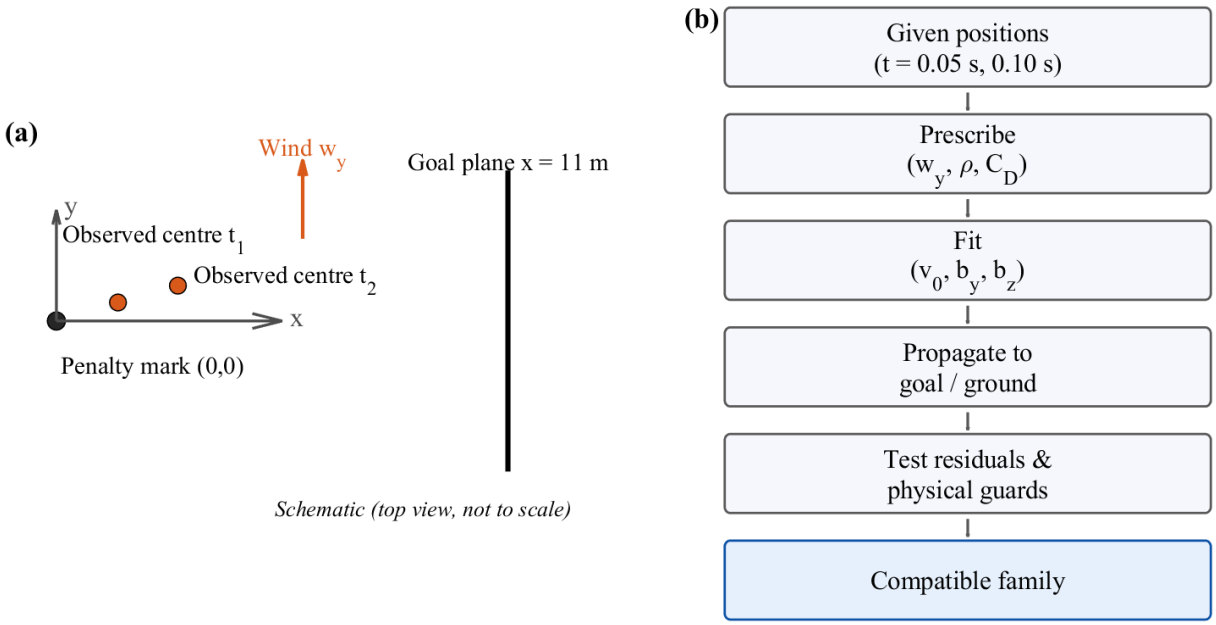


Figure 1: Coordinate convention and inverse-prediction workflow. The left panel is a horizontal schematic; the right panel shows how prescribed environments, inferred launch conditions and acceptance checks produce the compatible family.

2 Physical model and justified scope

2.1 Notation and prescribed information

Table 1: Notation used in the inverse problem. Prescribed scenario inputs and inferred quantities are distinguished.

Symbol	Meaning	Unit or role
$\mathbf{r}, \mathbf{v}$	Ball centre and velocity	m; m s^{-1}
$\mathbf{r}_i, t_i$	Two supplied positions and times	measured inputs
$\mathbf{u} = \mathbf{v} - \mathbf{w}$	Velocity relative to air	m s^{-1}
$\eta = (w_y, \rho, C_D)$	Prescribed environment profile	scenario inputs
$q = (\mathbf{v}_0, b_y, b_z)$	Inferred launch and effective lift	m s^{-1}; s^{-1}
ϵ	Coordinatewise observation tolerance	m
$C_\perp$	Effective transverse-force coefficient	dimensionless
t_g, y_g, z_g	First goal-plane crossing	s; m; m
σ	Conditional Gaussian-noise scale	m; Appendix A

2.2 Equations and parameters

Let $\mathbf{u} = \mathbf{v} - \mathbf{w}$ be velocity relative to air. Newton's second law is approximated by

$$\dot{\mathbf{r}} = \mathbf{v}, \tag{2}$$

$$\dot{\mathbf{v}} = -g\mathbf{e}_z - KC_D|\mathbf{u}|\mathbf{u} + \mathbf{b} \times \mathbf{u}, \qquad K = \frac{\rho\pi R^2}{2m}. \tag{3}$$

The drag term follows the conventional quadratic force law. Wind-tunnel studies establish that football drag and lateral forces depend on surface structure, flow regime and rotation [2]. They support including these mechanisms, but do not calibrate the particular ball in this problem.

The effective Magnus vector $\mathbf{b}$ has units of inverse seconds. If the lift coefficient is approximately proportional to spin parameter $R\omega/|\mathbf{u}|$, the transverse acceleration has this cross-product form. Thus $\mathbf{b}$ combines spin and aerodynamic efficiency; it is not a measurement of angular velocity. Constant $\mathbf{b}$ and C_D are short-flight approximations, subsequently stressed rather than treated as universal laws.

Table 2: Inputs and assumptions. Search bounds are modelling choices, not measured population intervals.

Quantity	Value	Status
Initial centre and observations	As supplied	Problem data
Ball radius R	0.11 m	Ground-contact assumption
Ball mass m	0.43 kg	Representative choice
Air density ρ	1.225 kg m^{-3}	Environmental assumption
Gravity g	9.80665 m s^{-2}	Standard representative value
Drag coefficient C_D	$[0.05, 0.60]$	Search domain
b_y, b_z	$[-1.5, 1.5]$ s^{-1}	Search domain
Coordinate error σ	$2, 5, 10, 20$ mm	Separate noise scenarios

The chosen mass and radius are consistent with the IFAB ball specifications of 410–450 g and 68–70 cm circumference [3]. Density and gravity are assumptions rather than site measurements. Initial speed bounds are $v_x \in [10, 50]$, $v_y \in [-20, 20]$, $v_z \in [-5, 25]$ m s^{-1}.

In the reference model $\mathbf{w} = \mathbf{0}$ and $b_x = 0$, leaving

$$\boldsymbol{\theta} = (v_{0x}, v_{0y}, v_{0z}, C_D, b_y, b_z). \tag{4}$$

Fixing b_x reduces a weakly identifiable direction; it does not establish that real spin has zero longitudinal component. All three spin components would give at least seven unknowns from six observations. Likewise, trajectory data primarily identify the product KC_D, so independently varying mass, density and fitted drag can double-count the same uncertainty.

2.3 Scale and conservation checks

The fitted initial speed is about 32.6 m s^{-1}. Using an assumed viscosity 1.81×10^{-5} Pa s gives $\mathrm{Re} = 2\rho R|\mathbf{v}|/\mu \simeq 4.9 \times 10^5$. This is a consistency check on using an aerodynamic force model, not a calibration of its coefficients. Over $T \simeq 0.40$ s, the dimensionless changes $KC_DUT \simeq 0.26$, $|\mathbf{b}|T \simeq 0.15$ and $gT/U \simeq 0.12$ show that drag, curvature and gravity can all materially affect extrapolation.

In still air the mechanical energy satisfies

$$\frac{d}{dt}\left(\frac{1}{2}m|\mathbf{v}|^2 + mgz\right) = -mKC_D|\mathbf{v}|^3 \leq 0, \tag{5}$$

because $\mathbf{v} \cdot (\mathbf{b} \times \mathbf{v}) = 0$. This is the appropriate dissipative balance. Ground-frame mechanical energy need not decrease monotonically when moving air performs work.

3 Information limits and analytical baselines

3.1 Why interpolation is not validation

After the initial position is fixed, the data contain six scalar constraints. A sufficiently flexible six-parameter model can interpolate these observations exactly, leaving no residual degrees of freedom with which to estimate noise. Further unknowns, including three-dimensional spin, variable drag, wind, and camera bias, cannot all be inferred freely.

For a simple constructive demonstration, adding $\mathbf{c}\,t(t-t_1)(t-t_2)$ to a smooth trajectory preserves all three observed positions but changes later positions. Physical restrictions limit admissible $\mathbf{c}$; observations alone do not. We therefore use a reduced physical model and test departures from it separately.

3.2 Constant acceleration

With $h = 0.05$ s and $\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{a}t^2$, exact interpolation gives

$$\mathbf{v}_0 = \frac{4\mathbf{r}_1 - \mathbf{r}_2 - 3\mathbf{r}_0}{2h} = (30.4, 9.4, 6.9)\ \mathrm{m\,s^{-1}}, \tag{6}$$

$$\mathbf{a} = \frac{\mathbf{r}_2 - 2\mathbf{r}_1 + \mathbf{r}_0}{h^2} = (-16, -16, -12)\ \mathrm{m\,s^{-2}}. \tag{7}$$

The earliest positive solution of $30.4t - 8t^2 = 11$ is 0.405008 s, giving $(y, z) = (2.494824, 1.920367)$ m. These extra digits are retained only as a reproducibility check. This acceleration is an inferred short-time average, not a force law guaranteed to remain constant.

A gravity-only baseline estimates initial velocity by linear least squares after subtracting the known gravitational displacement. A drag-only baseline estimates the three velocity components and a drag coefficient. Their residuals and predictions are compared with the full model in the numerical results.

3.3 Short observation windows amplify error

Assume that $\mathbf{r}_0$ is exact and each coordinate of $\mathbf{r}_1, \mathbf{r}_2$ has independent error with standard deviation σ. The acceleration uncertainty is

$$\mathrm{sd}(a_j) = \frac{\sqrt{5}\,\sigma}{h^2}. \tag{8}$$

At $\sigma = 0.01$ m this is 8.94 $\mathrm{m\,s^{-2}}$. At a fixed time $t = 8h = 0.40$ s, quadratic interpolation has weights $(21, -48, 28)$, so its coordinate uncertainty is $\sqrt{48^2 + 28^2}\,\sigma = 55.57\sigma$. Goal-plane prediction also requires uncertain crossing time; propagating uncertainty at fixed time alone is insufficient.

4 Full-flight constrained trajectory family: primary method

4.1 Coupled parameters and observation constraints

The same force law acts from release, $t = 0$, until the first terminal event. We retain steady lateral wind $\mathbf{w} = (0, w_y, 0)$, constant density, constant drag coefficient and constant effective transverse coefficients, with $b_x = 0$. Gusts, spin decay, drag crisis and camera bias are excluded from this family; they require additional information. The later structural stress tests remain separate comparisons.

Let $\eta = (w_y, \rho, C_D)$ and $q = (v_{0x}, v_{0y}, v_{0z}, b_y, b_z)$. The ideal observation equations are

$$\mathbf{r}(t_i; q, \eta) = \mathbf{r}_i, \quad i = 1, 2. \tag{9}$$

We relax them to componentwise constraints,

$$\mathcal{F}_\epsilon = \{(q, \eta) \in \mathcal{B} : \max_{i,j} |r_j(t_i; q, \eta) - r_{ij}| \le \epsilon\}, \qquad \epsilon = 0.005 \text{ m}. \tag{10}$$

The initial centre remains exactly the problem's $(0, 0, 0.11)$ m. The 5 mm tolerance is a chosen bounded-error scenario, not an inferred camera accuracy or a Gaussian standard deviation. It would match half a centimetre rounding interval only if rounding were the sole error. Results are also screened at 2, 10 and 20 mm. Exact numerical interpolation can occur; it does not establish exact physical knowledge.

Changing η requires solving for q again. A stronger aerodynamic deceleration can be compensated by a different launch velocity and transverse coefficient while the observations remain acceptable. The ball's launch kinetic energy, $E_0 = m|\mathbf{v}_0|^2/2$, is a useful comparison of kicks. It is not the foot's force: that would additionally require contact duration and ball deformation. The full problem has eight scalar unknowns and six measured coordinates. Moreover, only the product ρC_D appears in the drag term, so these two quantities cannot be separately identified in this reduced model. Effective b_y, b_z already absorb lift efficiency and spin; we do not also multiply them by an independently varied density factor.

4.2 Reproducible sampling of compatible solutions

We use seven equally spaced w_y values on $[-8, 8]$ m $\mathrm{s^{-1}}$, three densities $\{1.1, 1.225, 1.35\}$ kg $\mathrm{m^{-3}}$, and eleven drag values: ten equally spaced points on $[0.15, 0.60]$ plus the calm-air fitted value. These are illustrative search settings, not measured wind or ball distributions. For each of the 231 profiles, bounded nonlinear least squares estimates q from two starts:

$$q^*(\eta) = \arg\min_{q \in \mathcal{B}_q} \sum_{i=1}^{2} \|\mathbf{r}(t_i; q, \eta) - \mathbf{r}_i\|^2. \tag{11}$$

The effective transverse-coefficient bounds follow the reference model; launch bounds are tightened by the admissibility checks below. We evaluate the residual during fitting with RK4, step 0.0025 s, and independently re-evaluate acceptance with `ode45` from $t = 0$ (RelTol 10^{-11}, AbsTol 10^{-12}). Whole-flight propagation uses RelTol 10^{-9}, AbsTol 10^{-11} and maximum step 0.002 s. A release-contact guard avoids reporting the initial ground contact as a landing; the descending ground event and forward goal event are both localized.

We retain a profile representative only when every coordinate residual is within tolerance. Solver failures and rejected representatives are recorded. A failed least-squares representative is not proof that its entire profile has no feasible point: an L_2 minimizer need not minimize the largest residual. Likewise, two starts do not establish global optimality. Our displayed family is a sampled *inner subset* of $\mathcal{F}_\epsilon$, with one representative per sampled profile; it is neither a full feasible envelope nor a confidence region. We do not attach probability weights, and a fraction of candidates crossing the opening is not a scoring probability.

4.3 Physical admissibility is distinct from observation fit

A small residual is necessary but insufficient. We additionally require a forward, initially non-descending launch, $10 \le |\mathbf{v}_0| \le 40$ m s^{-1}, and compliance with the declared environment domain. Optimizer component bounds are tightened to $v_{0x} \in [10, 40]$ and $v_{0z} \in [0, 25]$ m s^{-1}; v_{0y} and b_y, b_z retain their previous bounds. IFAB specifies mass 0.410–0.450 kg and circumference 0.68–0.70 m [3]; the fixed 0.43 kg, 0.11 m radius satisfy these requirements. A study of nine experienced players observed ball speeds of 19.2–32.6 m s^{-1} [7]. The adopted 40 m s^{-1} cap includes a margin and is an explicit plausibility choice, not a universal human speed limit. No unsupported numerical upper bound on launch angle is imposed.

To prevent unrealistically large effective side forces, we screen

$$C_\perp(t) = \frac{\|\mathbf{b} \times \mathbf{u}(t)\|}{K\|\mathbf{u}(t)\|^2} \le 0.50. \tag{12}$$

This converts the fitted coefficient to a dimensionless transverse-force measure rather than pretending to recover true spin. Football wind-tunnel experiments provide the force-coefficient definition and show dependence on rotation [2]. The 0.50 cap and broad $C_D \in [0.15, 0.60]$ domain are conservative modelling guards, not experimentally established limits for every football. Density and wind ranges remain unmeasured scenario bounds. The force check samples the dense solution at spacings no larger than 0.001 s; it is a numerical admissibility check, not a proof of a continuous-time bound.

Candidates must also remain above the ground before their terminal event, survive beyond the last observation and produce a localized terminal event within 2 s. We do *not* constrain the inferred crossing to lie inside the goal: misses are legitimate outcomes. Observational compatibility, physical-screen acceptance and solver status are stored separately. A guard violation can indicate missing constraints, weak identifiability, a restricted search or a mismatch of assumptions; it does not by itself falsify Newton's equations.

At the default tolerance the maximum transverse coefficient among accepted representatives is 0.4538. All 107 pass the 0.50 guard. Tightening only this guard to 0.40 retains 98, and to 0.30 retains 66; relaxing to 0.60 retains 107. Speed caps 35, 40 and 45 m s^{-1} give the same counts on this fixed sample. This is screening sensitivity, not a reoptimized global feasible envelope. Rejected candidates and reasons remain in the output.

4.4 Computed family and coupled-kick example

All 231 optimizations converged without parameter-bound hits. At 5 mm, 107 representatives pass the six-coordinate test. They all reach the goal plane before the ground and clear the finite-radius opening in this chosen set. Their launch speeds span 32.323–32.849 m s^{-1}. Plane-coordinate ranges are $y \in [2.5342, 2.6812]$ m and $z \in [1.9451, 1.9867]$ m. The separate extrema need not belong to the same shot, and these sampled ranges do not bound untested solutions.

Table 3: Tolerance screening of the same profile representatives. Ranges are sampled spreads, not uncertainty bounds.

ϵ (mm)	Accepted	y_g (m)	z_g (m)
2	51/231	[2.570,2.623]	[1.955,1.975]
5	107/231	[2.534,2.681]	[1.945,1.987]
10	189/231	[2.445,2.756]	[1.918,2.005]
20	231/231	[2.280,2.815]	[1.885,2.013]

Table 4: Two compatible full-flight solutions, selected for largest pairwise separation within the 5 mm sample. Wind is present from release in both.

	w_y	ρ	C_D	$\lvert v_0 \rvert$	Error (mm)	y_g	z_g
A	-8.0	1.225	0.2500	32.323	4.834	2.6812	1.9847
B	-8.0	1.350	0.3707	32.791	3.828	2.5342	1.9492

Units: wind and speed in m s^{-1}; density in kg m^{-3}; plane coordinates in m.

The two plane crossings differ by 0.1513 m despite both meeting the same observation tolerance. Their residuals, velocity vectors, effective force coefficients and complete trajectories are retained with the numerical output. This demonstrates non-uniqueness within the stated physical family without inventing an unobserved force switch. The observed spread is modest because these are steady, reduced-force profiles refitted to the observations; complexity alone would not justify a larger or more accurate result.

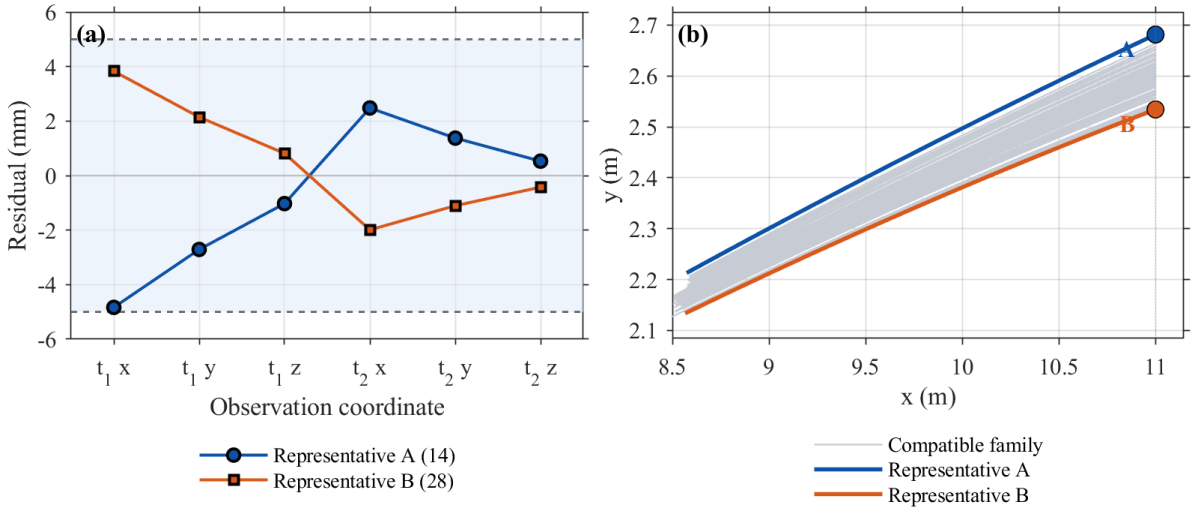


Figure 2: The same two candidate shots satisfy all six coordinate tolerances (left), while their late-flight paths separate in the horizontal plane (right). The shaded residual band is the prescribed 5 mm tolerance. Gray paths are compatible representatives; the actual displacements are not rescaled.

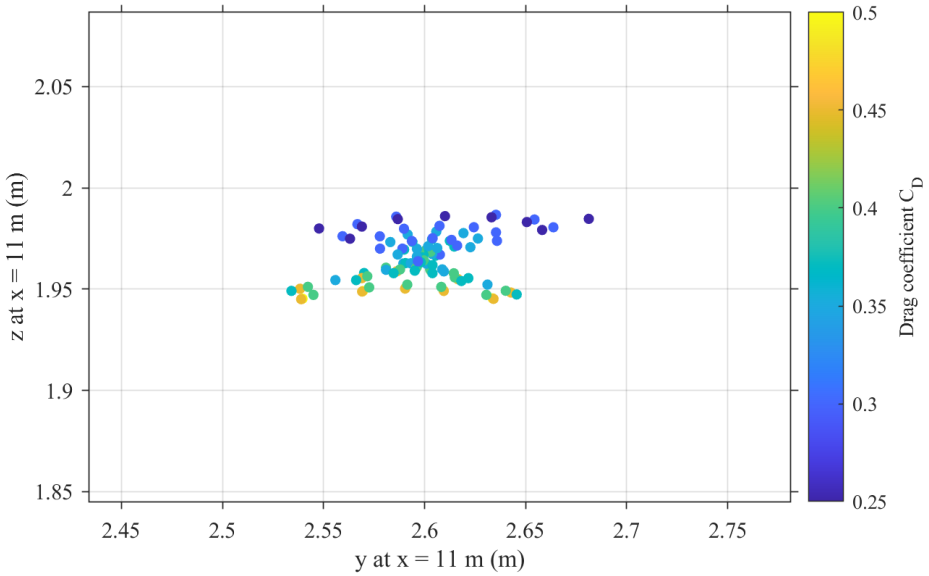


Figure 3: Zoomed goal-plane scatter for compatible profile representatives; color indicates prescribed C_D.

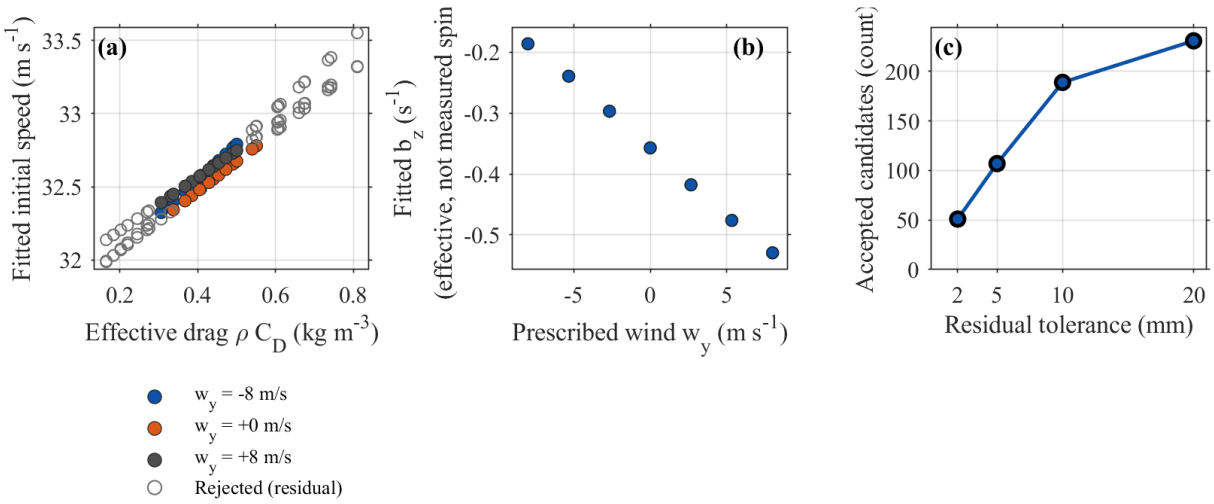


Figure 4: Parameter coupling and screening: prescribed density–drag product versus inferred launch speed for three wind slices (left), wind versus effective b_z for compatible representatives (middle), and accepted profile count versus coordinate tolerance (right). Counts are properties of this sampled search.

4.5 Numerical evidence and interpretation of "most likely"

Python independently integrated all 231 candidates using DOP853 and reproduced the acceptance decisions without disagreements; maximum plane-point difference from MATLAB was $1e-13$ m. Independent refits of the two displayed representatives also agreed. Tests additionally verify wind at release, compensating initial-velocity changes, invariance under fixed ρC_D, three maximum-step settings, drag dissipation relative to air, transverse-force orthogonality, ground-event localization and the ground-frame work balance

$$\frac{d}{dt}\left(\frac{1}{2}m|\mathbf{v}|^2 + mgz\right) = (\mathbf{F}_D + \mathbf{F}_M)\cdot\mathbf{v}. \tag{13}$$

These are numerical checks, not experimental confirmation. The conditional calm-air model and synthetic tests in the appendices provide additional diagnostics; their Gaussian confidence estimates must not be substituted for the bounded-error family's sampled spread.

Without a measurement likelihood and a probability model for the unknown environment, these data do not specify an unconditional most-likely crossing. To give the requested point estimate, we retain the calm-air maximum-likelihood reference $(11, 2.60, 1.97)$ m at 0.401 s, explicitly conditional on its assumptions, alongside the compatible family. A predictive mode marginalized over nuisance parameters is not identified by these two observations alone. The interactive MATLAB view implements the same inverse model: releasing a parameter slider recomputes the entire shot and restarts its animation, while reporting fitted launch conditions and residual acceptance. The interactive exploration ranges are $w_y \in [-15, 15]$ m s^{-1}, $\rho \in [0.90, 1.40]$ kg m^{-3} and $C_D \in [0.08, 0.80]$. These broader stress-test controls are screened by the same launch-speed and force checks; the reported family statistics use the prescribed grid stated above.

5 Operational meaning, limitations and improvements

The nominal centre lies on the positive-y side of the goal, high enough that vertical uncertainty matters. At 0.10 s, only about 0.30 s remains before the predicted crossing. The precision implied by the input decimal places must not be confused with known camera accuracy: rounding to centimetres alone would be a different error model from independent Gaussian noise of 1 cm standard deviation.

The leading limitation is the short measurement interval. Additional frames could distinguish curvature changes and reduce velocity–acceleration coupling; calibrated camera error covariance and synchronized timestamps would make confidence estimates more defensible. Common camera bias, uncertain initial position, depth reconstruction and correlated coordinate errors are not captured by the independent-error scenario. For general known covariance Σ, replace J^TJ/σ^2 by $J^T\Sigma^{-1}J$.

Constant effective Magnus force structure cannot resolve non-stationary wakes, abrupt gusts or force changes after the last observation. No data-driven finite bound on arbitrary future disturbances exists. Low-spin flutter and rebound require additional calibrated models. For trajectories that touch the ground before the goal, our method reports a different outcome instead of silently extrapolating below the surface.

Independent implementations and synthetic validation check equations and algorithms. They do not establish that this particular football, field and camera obey the chosen model. There is no independent measured full trajectory here; experimental confirmation remains outstanding. Bounds and synthetic sampling ranges are explicit scenario choices, not universal football statistics.

6 Conclusions

The primary answer is a coupled full-flight inverse problem: environmental parameters and launch conditions must be inferred together under the two observation constraints. At a stated 5 mm coordinate tolerance, 107 of 231 sampled profile representatives are compatible, and two differ by 0.1513 m at the goal. These ranges demonstrate non-uniqueness but are not global bounds or probabilities. The calm-air point $(11, 2.60, 1.97)$ m at 0.401 s remains a conditional reference, not a uniquely established truth or an unconditional predictive mode. Gaussian measurement-error diagnostics are reported separately. Further observations, environmental measurements and independent flight data are needed to narrow the feasible set and calibrate probabilistic forecasts.

A Conditional reference: likelihood and measurement diagnostics

A.1 Bounded maximum likelihood

For a chosen σ, we minimize

$$\chi^2(\boldsymbol{\theta}) = \sum_{i=1}^{2} \frac{\|\mathbf{r}(t_i;\boldsymbol{\theta}) - \mathbf{r}_i\|^2}{\sigma^2} \tag{14}$$

over the stated bounds using `lsqnonlin` [5]. Equal coordinate variances mean changing σ alone rescales the objective without changing an interior optimum. Multiple starting points check local consistency; they do not prove global optimality.

The point forecast is the goal-plane image of the constrained maximum-likelihood parameter estimate. In this precise sense it is our model-conditioned "most likely" forecast. A mode of a marginal landing-point probability density would additionally require a probabilistic parameter model and integration over nuisance parameters; we do not claim such an unconditional mode.

Observations are evaluated using fourth-order Runge–Kutta steps of 0.005 s. Subsequent flight uses `ode45` with directional events [4]: increasing $x - 11 = 0$ and decreasing $z - R = 0$. The same force law is used for observation propagation and event prediction. Missing crossings, numerical failures, and earlier ground contacts are recorded explicitly. Applicability assumes the ball is airborne through the observation window.

The problem's centre-plane criterion is $|y| < 3.66$, $0 < z < 2.44$. A stricter opening clearance is $|y| < 3.66 - R$ and $R \le z < 2.44 - R$. We also compute the centre at $x = 11 + R$ as a finite-size sensitivity; this is an ideal thin-plane approximation, not a detailed post-collision model.

A.2 Local confidence region

Let $\mathbf{f}(\boldsymbol{\theta}) \in \mathbb{R}^6$ collect predicted observations, $J = \partial\mathbf{f}/\partial\boldsymbol{\theta}$, and $G = \partial(y_g, z_g)/\partial\boldsymbol{\theta}$, including variation of crossing time. For a full-rank interior solution,

$$C_\theta \simeq \sigma^2 (J^T J)^{-1}, \qquad C_g \simeq G C_\theta G^T. \tag{15}$$

The local 95% joint ellipse is

$$(\mathbf{q} - \widehat{\mathbf{q}})^T C_g^{-1} (\mathbf{q} - \widehat{\mathbf{q}}) \le 5.991. \tag{16}$$

Marginal coordinate half-widths are $1.959964\sqrt{(C_g)_{jj}}$. These are not jointly 95% rectangular bounds. Near active parameter constraints, Gaussian linearization is approximate and may misrepresent coverage.

A.3 Resampling and model uncertainty

Parametric bootstrap generates noisy observations from the fitted model, refits every sample and propagates each successful fit. We report boundary hits, failures, ground contacts and plane crossings alongside coordinate quantiles. This describes the estimator's sampling stability. The fraction of bootstrap forecasts within the opening is neither an experimentally established scoring probability nor a Bayesian posterior probability.

Structural tests change one mechanism at a time: fixed $b_x = \pm 1$ s^{-1}, transverse wind $w_y = \pm 3$ m s^{-1}, $C_D = C_{D0}(|\mathbf{u}|/30)^p$ with $p = \pm 0.5$, and $\mathbf{b}(t) = \mathbf{b}_0 e^{-t/\tau}$ with $\tau = 0.5, 1$ s. Each is refitted to the same observations. These deliberately chosen stress cases are not assigned occurrence probabilities and must not be pooled into a statistical confidence interval.

B Additional reference-model results and verification

B.1 Point forecast and competing baselines

The four independent starts converge to the same interior solution, with parameters

$$\widehat{\boldsymbol{\theta}} = (30.41546, 9.41672, 6.90801, 0.370687, -0.060326, -0.357128). \tag{17}$$

The first three entries have units m s^{-1}, the fourth is dimensionless, and the last two have units s^{-1}. The predicted crossing is $(y_g, z_g) = (2.599441, 1.967005)$ m at 0.4005035 s. This numerical precision permits reproducibility; the answer should be reported as $(11, 2.60, 1.97)$ m at 0.401 s. The drag-only objective is $\chi^2 = 1.517$ for two nominal residual degrees of freedom,

Table 5: Observation fit and extrapolation. Tiny interpolation residuals do not demonstrate physical validity.

Model	RMSE (mm)	t_g (s)	y_g (m)	z_g (m)
Gravity only	10.376	0.3706	3.217	1.957
Constant acceleration	0.000	0.4050	2.495	1.920
Drag only	5.029	0.4058	3.217	1.879
Drag and effective Magnus	0.000	0.4005	2.599	1.967

corresponding to an approximate goodness-of-fit tail probability 0.468 under the 1 cm Gaussian assumption. Thus these observations do not by themselves establish the need for the two extra transverse-force parameters at that noise level. At 2 mm noise, the same drag-only discrepancy gives $\chi^2 = 37.9$, making the interpretation strongly dependent on measurement quality. The more flexible model is retained as a conditional curvature-aware forecast, not declared uniquely correct by its exact fit.

At the centre plane $x = 11.11$ m, the finite-radius sensitivity gives $(y, z) = (2.61689, 1.97822)$ m at 0.40493 s. The additional travel shifts the centre by approximately 2.1 cm in the goal plane. The nominal shot clears the ideal opening under both conventions.

B.2 Measurement uncertainty and bootstrap stability

For independent coordinate errors of standard deviation 0.01 m, the local prediction covariance is

$$C_g \simeq \begin{pmatrix} 0.245132 & 0.000117 \\ 0.000117 & 0.243555 \end{pmatrix} \text{ m}^2. \tag{18}$$

The marginal 95% half-widths are 0.9704 m and 0.9673 m. At noise levels of 2, 5 and 20 mm, the respective half-widths scale to approximately 0.194, 0.485 and 1.94 m in each coordinate. This scaling is a local result; it is not a claim that bounded nonlinear resampling scales exactly at every noise level. The 95% joint ellipse has semiaxes 1.212 and 1.208 m. The unscaled observation Jacobian condition number is 15.0; it depends on parameter units and is reported as a numerical diagnostic, not an invariant identifiability score.

All 600 bootstrap fits terminate successfully; 53 hit at least one parameter boundary. All reach the goal plane, but 165 fail the finite-radius opening-clearance criterion. The latter are retained in the coordinate summaries and figure. The empirical 2.5–97.5 percentile ranges of the re-estimated plane coordinates are $[1.703, 3.617]$ m in y and $[0.967, 2.937]$ m in z. These are estimator resampling quantiles, not a posterior distribution for the true shot. The 72.5% within-opening frequency is a stability diagnostic only. Boundary hits and the region extending beyond the frame are material features of the result.

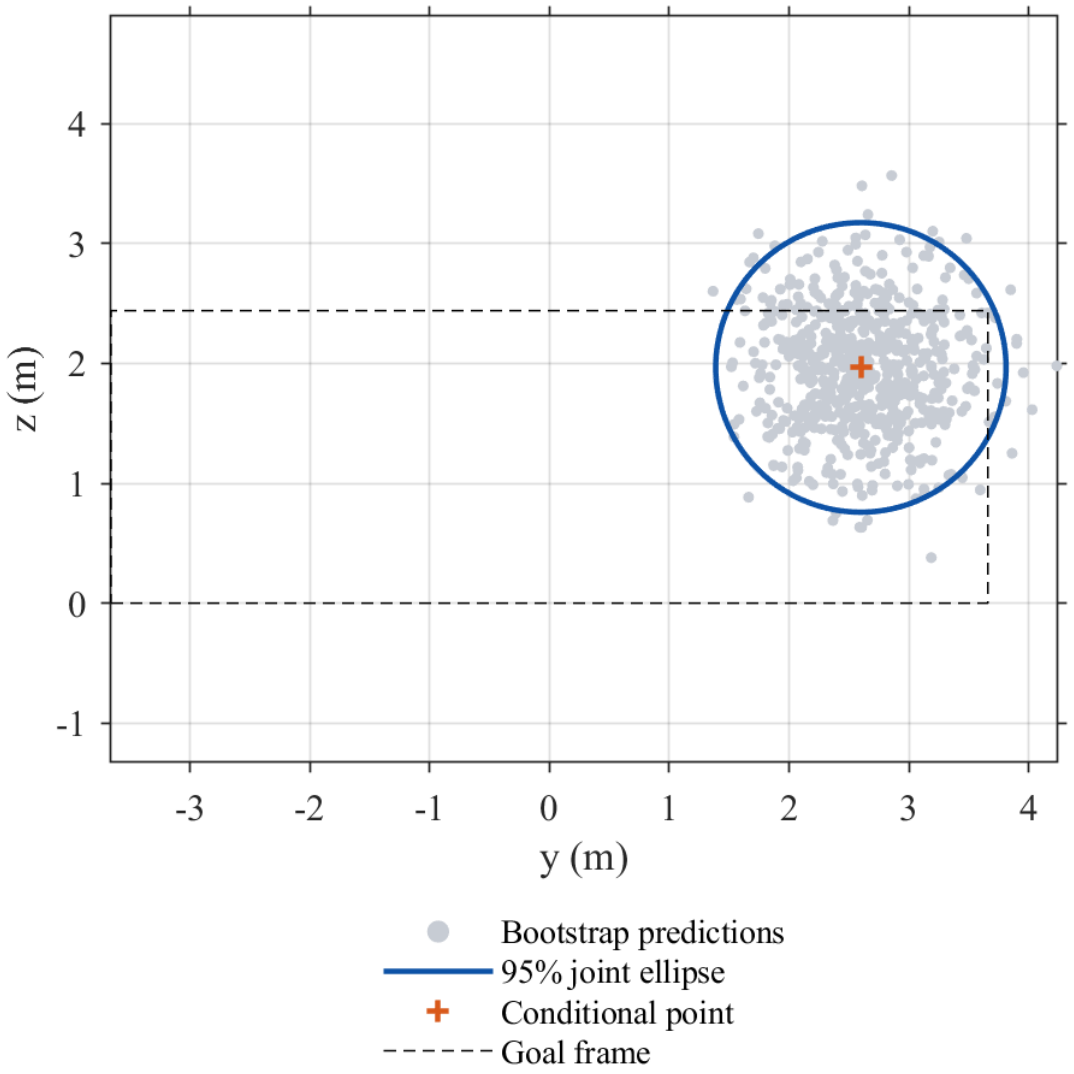


Figure 5: Goal-plane forecasts from 600 bootstrap refits with 1 cm coordinate noise. All plane crossings are retained, including forecasts outside the opening. The ellipse is the local 95% joint confidence approximation.

B.3 Verification and synthetic forecast testing

MATLAB R2024a performs the main integration, optimization and resampling, with random seed 20261107. The independent Python implementation uses SciPy 1.17.1, DOP853 integration and bounded nonlinear least squares. It agrees with the nominal MATLAB goal-plane coordinates to less than 2×10^{-11} m. This very small difference establishes implementation consistency, not real-world prediction accuracy. The RK4 gravity benchmark has maximum position error $2.7e-15$ m and the one-dimensional quadratic-drag benchmark $5.2e-13$ m. At fixed fitted parameters, step sizes 0.01, 0.005 and 0.0025 s change the predicted crossing by less than 3×10^{-10} m. Independently changing the event integrator's maximum step to 0.02, 0.01 and 0.005 s changes the result by less than 4×10^{-12} m. Tolerances are RelTol=10^{-9} and AbsTol=10^{-11} for event integration. Refinement checks concern integration error, not uncertainty in the fitted physics. No convergence order is claimed from roundoff-scale errors. An independent integrated energy check balances mechanical-energy loss against drag dissipation to 2×10^{-13} J.

For the synthetic tests, each of 150 truth vectors is drawn independently and uniformly over $v_x \in [24, 34]$, $v_y \in [3, 10]$, $v_z \in [4, 9]$ m s^{-1}, $C_D \in [0.15, 0.40]$, $b_y \in [-0.15, 0.15]$, $b_z \in [-0.60, 0.10]$ s^{-1}. These ranges define a designed test set, not observed population statistics. High-accuracy ode45 trajectories generate the early observations, independently of the RK4 fitting routine. Noise of standard deviation 1 cm is then added and each case is refitted from two starts.

All 150 inversions converge, with 31 boundary hits. All true trajectories reach the plane; 3 fitted trajectories instead hit the ground first. For the 147 paired plane crossings, two-coordinate RMSE is 0.727 m. The local 95% ellipse covers 146 of those 147 true crossings (99.32%); counting the three non-crossing forecasts as noncoverage yields 146/150 (97.33%). Neither number proves exact 95% calibration, especially with only 150 cases and active constraints. Opening/outside/ground classification agrees in 123/150 cases. Synthetic data obey the reference force family, so these results do not test structural correctness against real ball flight.

B.4 Structural stress cases

Table 6: One-factor changes, each refitted to the same observations. Distances are relative to the nominal crossing; these are scenarios, not confidence bounds.

Scenario	y_g (m)	z_g (m)	Shift (cm)
bx=-1	2.5516	2.0402	8.75
bx=+1	2.6601	1.9037	8.77
wind_y=-3	2.6004	1.9669	0.10
wind_y=+3	2.5983	1.9664	0.13
Cd exponent p=-0.5	2.5983	1.9648	0.25
Cd exponent p=+0.5	2.6005	1.9690	0.22
b decay tau=0.5 s	2.7135	1.9503	11.53
b decay tau=1.0 s	2.6611	1.9579	6.23

Each stress case fits the early positions to numerical precision, yet future positions differ. The largest shift in this chosen set is about 11.5 cm for the 0.5 s decay case. Steady transverse wind has a small residual effect after refitting because the free initial velocity and force parameters absorb much of its early influence; this does not imply that an unexpected later gust is harmless. Unobserved longitudinal spin causes roughly 9 cm shifts. The stress range is illustrative and does not bound all physically possible future disturbances.

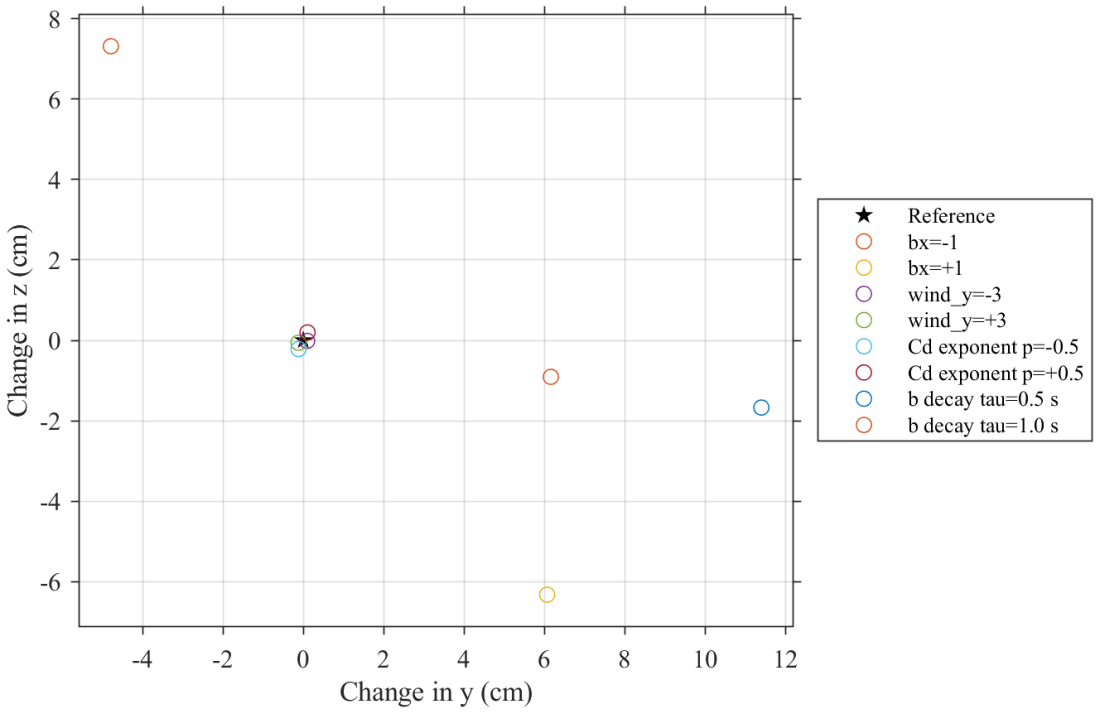


Figure 6: Displacement from the nominal goal-plane forecast after one-factor refitting. Units are centimetres; the selected scenarios carry no probability weights.

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Report on the Use of AI

OpenAI Codex assisted with problem selection, modelling decisions, source checking, independent numerical verification and manuscript preparation. Claude Code, routed through the user's existing provider to DeepSeek `deepseek-v4-flash`, assisted with MATLAB implementation. Human team members must verify and take responsibility for all adopted content.

Draft disclosure: not ready for submission. The official rules require the exact input and complete output of AI tools [6]. This paragraph is a role summary, not a substitute for those records. The complete adopted conversation and execution transcripts must be appended and reviewed before submission; team number is also awaiting confirmation. No contest materials have been submitted automatically.